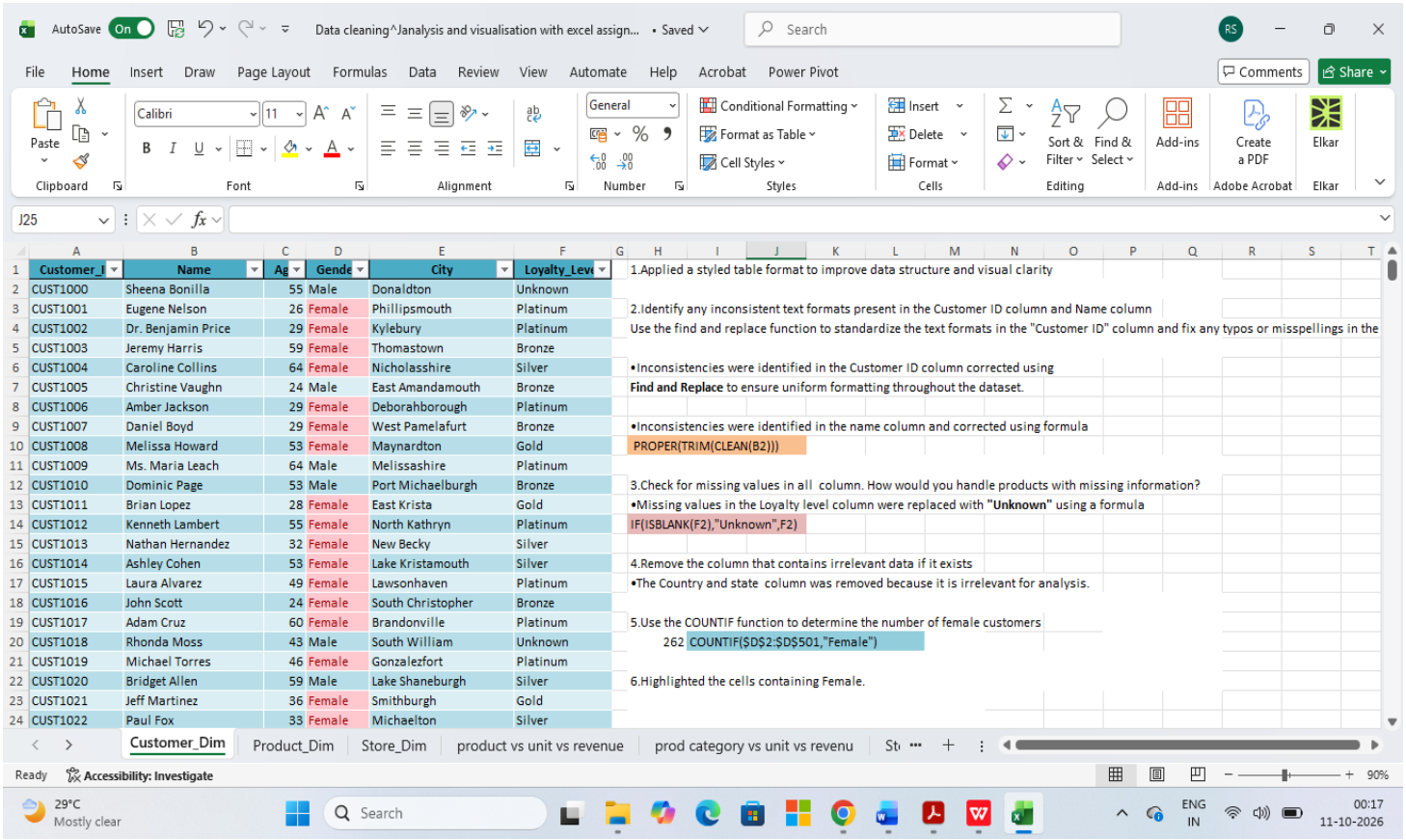
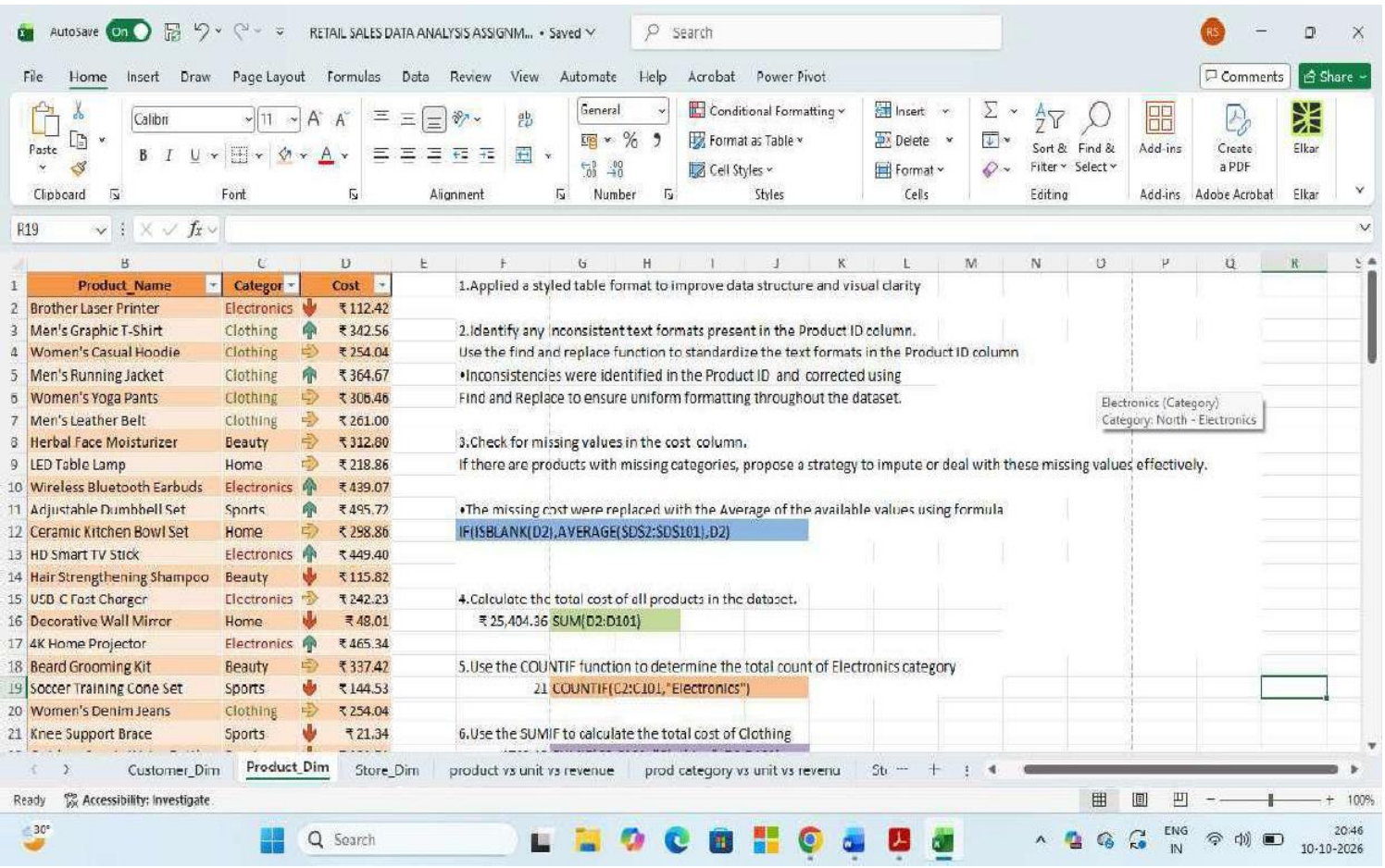
ASSIGNMENT TITLE: DATA CLEANING, ANALYSIS & VISUALIZATION WITH EXCEL

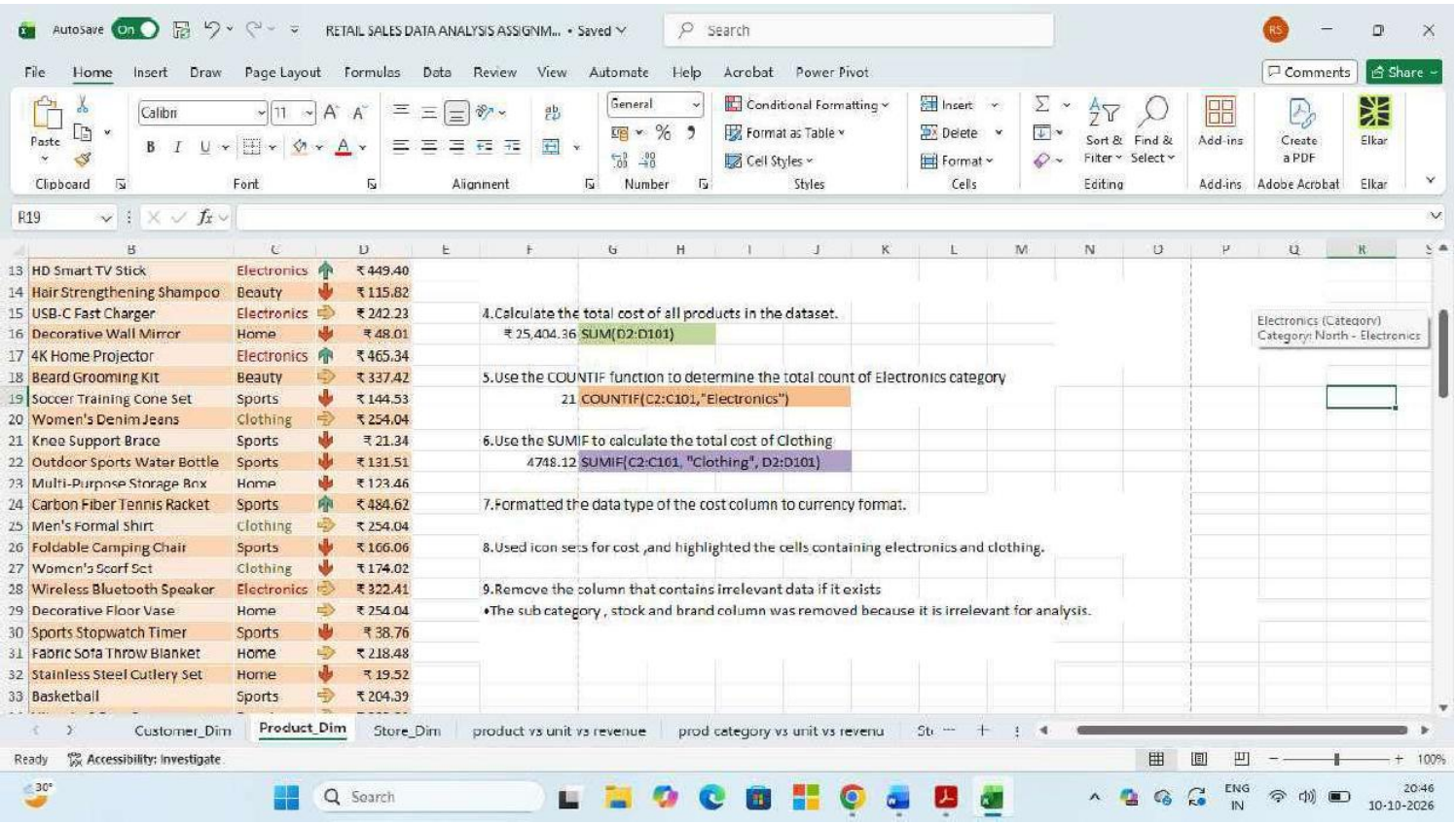
PROJECT TOPIC: E-COMMERCE SALES DATA ANALYSIS USING MICROSOFT EXCEL

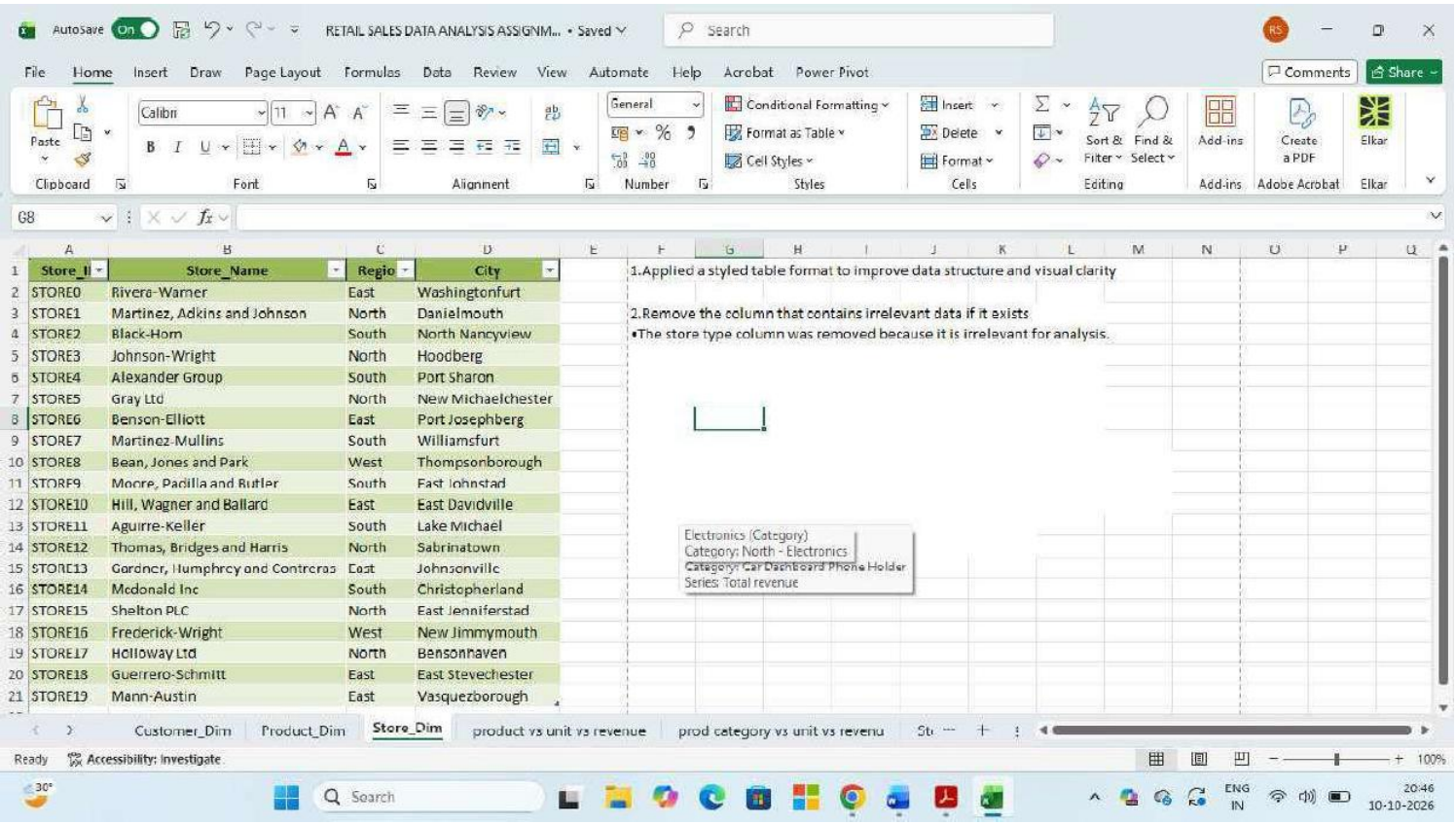
Summary:

- This project focuses on cleaning, analysing, and visualizing an E-commerce sales dataset using Microsoft Excel.
- It involves identifying duplicate records, handling missing and inconsistent data, applying Excel formulas and lookup functions, and summarizing information using Pivot Tables and descriptive statistics.
- Charts, Pivot Charts, and an interactive dashboard are developed to present key sales insights and support data-driven business decisions.
- This assignment demonstrates the practical application of Excel for data analysis, business reporting, and visualization as part of my journey toward becoming an aspiring Data Analyst.

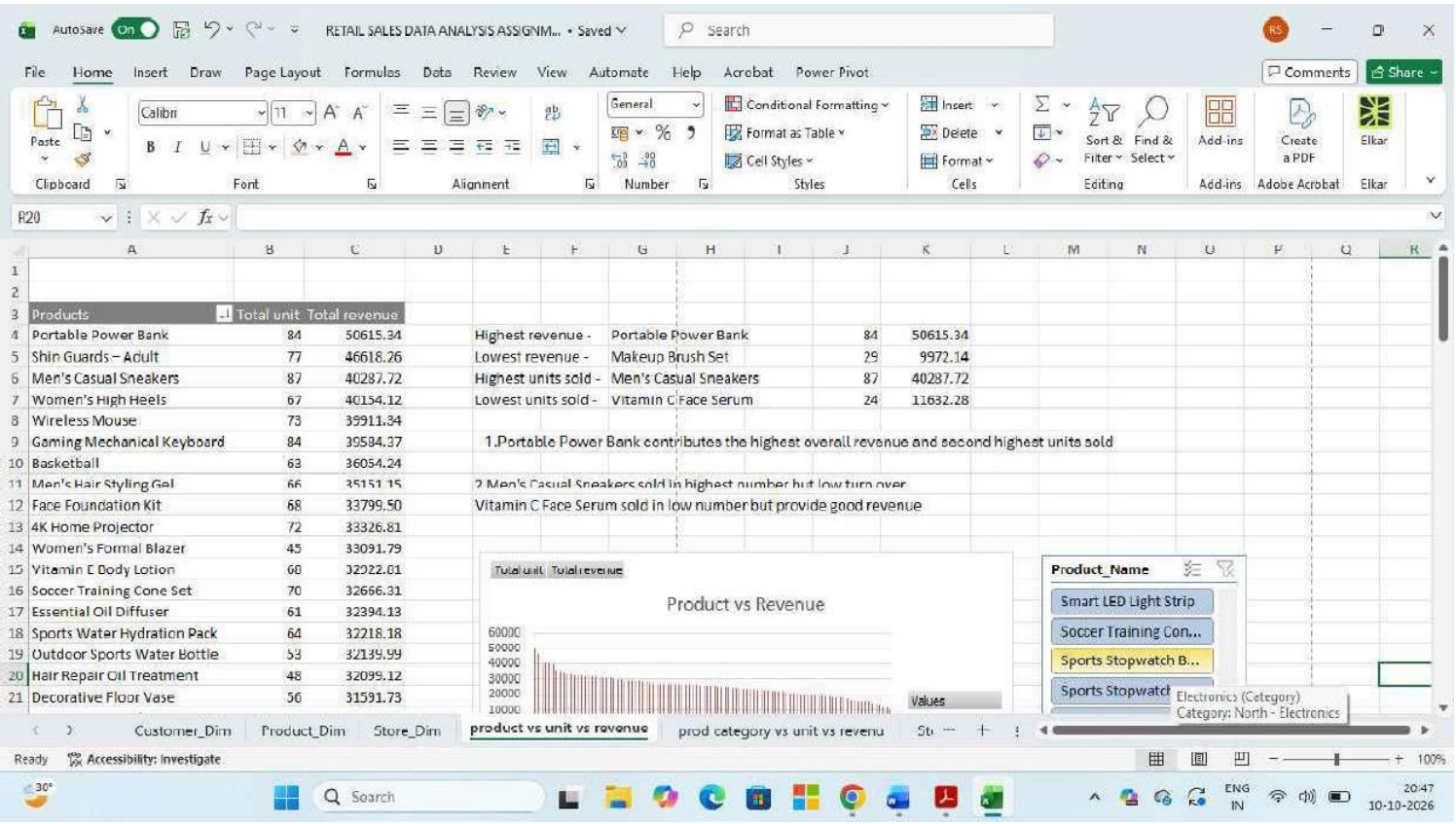


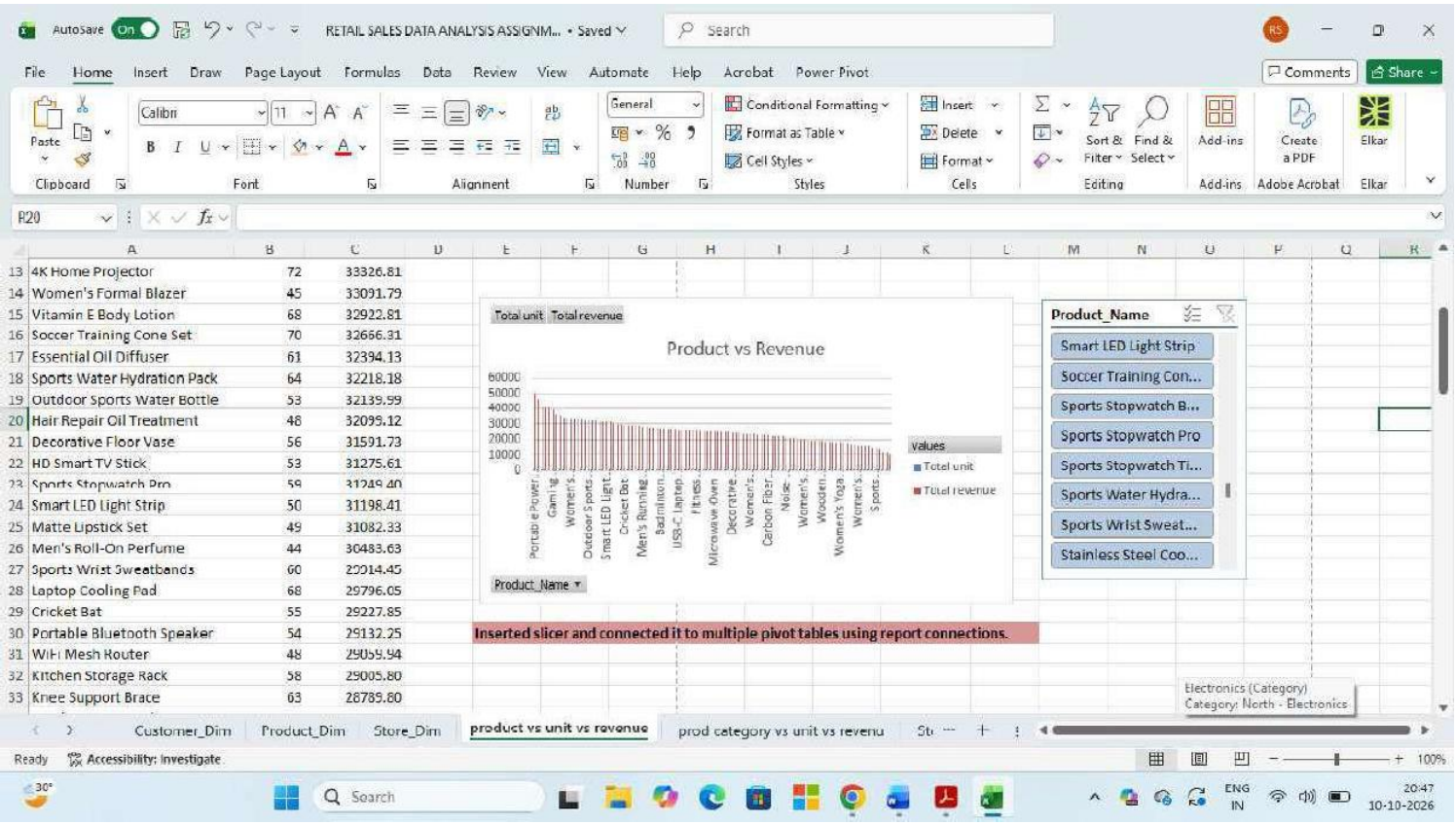

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Calibri 11 General Conditional Formatting Format as Table Cell Styles Insert Delete Format Sort & Filter Find & Select Add-ins Create a PDF Elkar
Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
R19
Product_Name Categor Cost
Brother Laser Printer Electronics ₹112.42
Men's Graphic T-Shirt Clothing ₹342.56
Women's Casual Hoodie Clothing ₹254.04
Men's Running Jacket Clothing ₹364.67
Women's Yoga Pants Clothing ₹306.46
Men's Leather Belt Clothing ₹261.00
Herbal Face Moisturizer Beauty ₹312.80
LED Table Lamp Home ₹218.86
Wireless Bluetooth Earbuds Electronics ₹439.07
Adjustable Dumbbell Set Sports ₹495.72
Ceramic Kitchen Bowl Set Home ₹298.86
HD Smart TV Stick Electronics ₹449.40
Hair Strengthening Shampoo Beauty ₹115.82
USB C Fast Charger Electronics ₹242.23
Decorative Wall Mirror Home ₹48.01
4K Home Projector Electronics ₹465.34
Beard Grooming Kit Beauty ₹337.42
Soccer Training Cone Set Sports ₹144.53
Women's Denim Jeans Clothing ₹254.04
Knee Support Brace Sports ₹21.34
1.Applied a styled table format to improve data structure and visual clarity
2.Identify any inconsistent text formats present in the Product ID column.
Use the find and replace function to standardize the text formats in the Product ID column
•Inconsistencies were identified in the Product ID and corrected using
Find and Replace to ensure uniform formatting throughout the dataset.
Electronics (Category)
Category: North - Electronics
3.Check for missing values in the cost column.
If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.
•The missing cost were replaced with the Average of the available values using formula
IF(ISBLANK(D2),AVERAGE($D$2:$D$101),D2)
4.Calculate the total cost of all products in the dataset.
₹ 25,404.36 SUM(D2:D101)
5.Use the COUNTIF function to determine the total count of Electronics category
21 COUNTIF(C2:C101,"Electronics")
6.Use the SUMIF to calculate the total cost of Clothing
Customer_Dim Product_Dim Store_Dim product vs unit vs revenue prod category vs unit vs revenu
Ready Accessibility: Investigate
100%
Search
ENG IN
20:46
10-10-2026

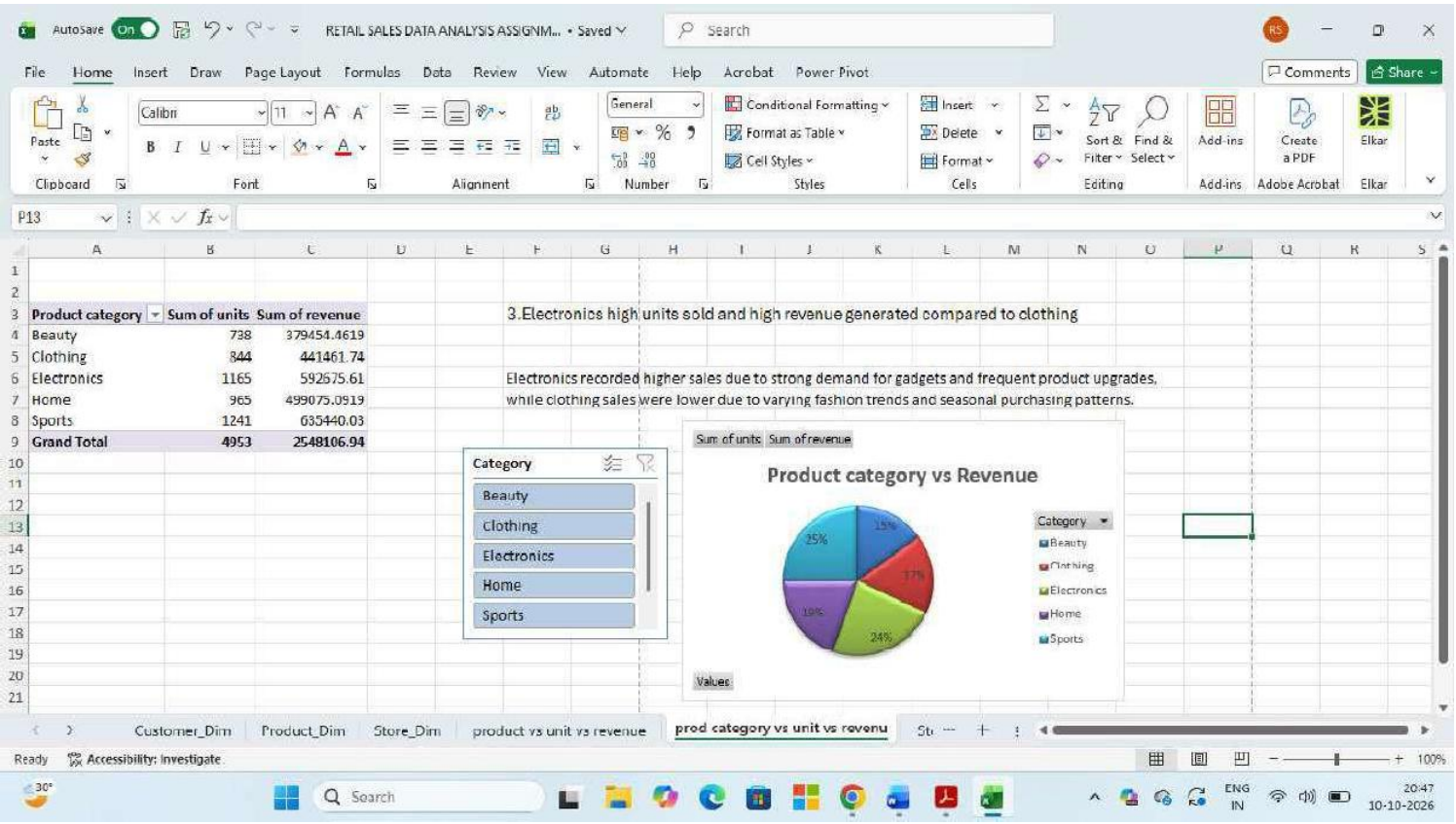

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Calibri 11 Font Alignment General Number Conditional Formatting Format as Table Cell Styles Styles Insert Delete Format Cells Sort & Filter Find & Select Editing Add-ins Add-ins Create a PDF Adobe Acrobat Elkar Elkar
R19
HD Smart TV Stick Electronics ₹449.40
Hair Strengthening Shampoo Beauty ₹115.82
USB-C Fast Charger Electronics ₹242.23
Decorative Wall Mirror Home ₹48.01
4K Home Projector Electronics ₹465.34
Beard Grooming Kit Beauty ₹337.42
Soccer Training Cone Set Sports ₹144.53
Women's Denim Jeans Clothing ₹254.04
Knee Support Brace Sports ₹21.34
Outdoor Sports Water Bottle Sports ₹131.51
Multi-Purpose Storage Box Home ₹123.46
Carbon Fiber Tennis Racket Sports ₹484.62
Men's Formal Shirt Clothing ₹254.04
Foldable Camping Chair Sports ₹166.06
Women's Scarf Set Clothing ₹174.02
Wireless Bluetooth Speaker Electronics ₹322.41
Decorative Floor Vase Home ₹254.04
Sports Stopwatch Timer Sports ₹38.76
Fabric Sofa Throw Blanket Home ₹218.48
Stainless Steel Cutlery Set Home ₹19.52
Basketball Sports ₹204.39
4.Calculate the total cost of all products in the dataset.
₹ 25,404.36 SUM(D2:D101)
5.Use the COUNTIF function to determine the total count of Electronics category
21 COUNTIF(C2:C101,"Electronics")
6.Use the SUMIF to calculate the total cost of Clothing
4748.12 SUMIF(C2:C101, "Clothing", D2:D101)
7.Formatted the data type of the cost column to currency format.
8.Used icon sets for cost ,and highlighted the cells containing electronics and clothing.
9.Remove the column that contains irrelevant data if it exists
•The sub category , stock and brand column was removed because it is irrelevant for analysis.
Electronics (Category)
Category: North - Electronics
Customer_Dim Product_Dim Store_Dim product vs unit vs revenue prod category vs unit vs revenu
Ready Accessibility: Investigate
100%
Search
ENG IN
20:46
10-10-2026

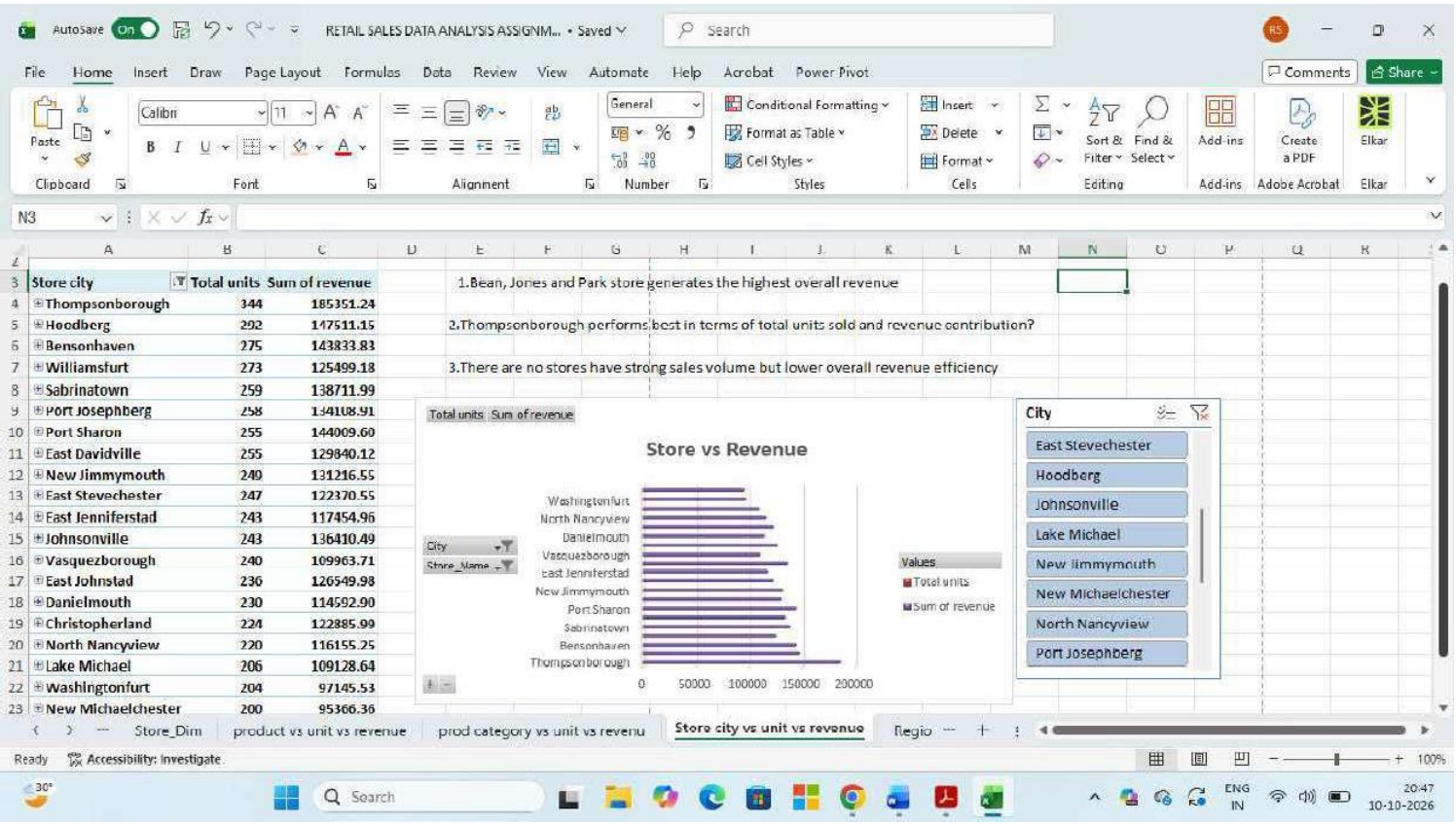


Store_ID	Store_Name	Region	City
STORE0	Rivera-Warner	East	Washingtonfurt
STORE1	Martinez, Adkins and Johnson	North	Danielmouth
STORE2	Black-Horn	South	North Nancyview
STORE3	Johnson-Wright	North	Hoodberg
STORE4	Alexander Group	South	Port Sharon
STORE5	Gray Ltd	North	New Michaelchester
STORE6	Benson-Elliott	East	Port Josephberg
STORE7	Martinez-Mullins	South	Williamsfurt
STORE8	Bean, Jones and Park	West	Thompsonborough
STORE9	Moore, Padilla and Butler	South	East Johnstad
STORE10	Hill, Wagner and Ballard	East	East Davidville
STORE11	Aguirre-Keller	South	Lake Michael
STORE12	Thomas, Bridges and Harris	North	Sabrinatown
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville
STORE14	Mcdonald Inc	South	Christopherland
STORE15	Shelton PLC	North	East Jenniferstad
STORE16	Frederick-Wright	West	New Jimmymouth
STORE17	Holloway Ltd	North	Bensonhaven
STORE18	Guerrero-Schmitt	East	East Stevechester
STORE19	Mann-Austin	East	Vasquezborough


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Sort & Filter
Find & Select
Create a PDF
R20
Products Total unit Total revenue
Portable Power Bank 84 50615.34
Shin Guards – Adult 77 46618.26
Men's Casual Sneakers 87 40287.72
Women's High Heels 67 40154.12
Wireless Mouse 73 39911.34
Gaming Mechanical Keyboard 84 39584.37
Basketball 63 36054.24
Men's Hair Styling Gel 66 35151.15
Face Foundation Kit 68 33799.50
4K Home Projector 72 33326.81
Women's Formal Blazer 45 33091.79
Vitamin E Body Lotion 60 32922.01
Soccer Training Cone Set 70 32666.31
Essential Oil Diffuser 61 32394.13
Sports Water Hydration Pack 64 32218.18
Outdoor Sports Water Bottle 53 32139.99
Hair Repair Oil Treatment 48 32099.12
Decorative Floor Vase 56 31591.73
Highest revenue - Portable Power Bank 84 50615.34
Lowest revenue - Makeup Brush Set 29 9972.14
Highest units sold - Men's Casual Sneakers 87 40287.72
Lowest units sold - Vitamin C Face Serum 24 11632.28
1.Portable Power Bank contributes the highest overall revenue and second highest units sold
2.Men's Casual Sneakers sold in highest number but low turn over
Vitamin C Face Serum sold in low number but provide good revenue
Total unit Total revenue
Product vs Revenue
60000
50000
40000
30000
20000
10000
Values
Product_Name
Smart LED Light Strip
Soccer Training Con...
Sports Stopwatch B...
Sports Stopwatch
Electronics (Category)
Category: North - Electronics
Customer_Dim Product_Dim Store_Dim product vs unit vs revenue prod category vs unit vs revenu
Ready Accessibility: Investigate
100%
Search
ENG IN
20:47
10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Sort & Filter
Find & Select
Create a PDF
R20
4K Home Projector 72 33326.81
Women's Formal Blazer 45 33091.79
Vitamin E Body Lotion 68 32922.81
Soccer Training Cone Set 70 32666.31
Essential Oil Diffuser 61 32394.13
Sports Water Hydration Pack 64 32218.18
Outdoor Sports Water Bottle 53 32139.99
Hair Repair Oil Treatment 48 32095.12
Decorative Floor Vase 56 31591.73
HD Smart TV Stick 53 31275.61
Sports Stopwatch Pro 59 31249.40
Smart LED Light Strip 50 31198.41
Matte Lipstick Set 49 31082.33
Men's Roll-On Perfume 44 30483.63
Sports Wrist Sweatbands 60 29914.45
Laptop Cooling Pad 68 29796.05
Cricket Bat 55 29227.85
Portable Bluetooth Speaker 54 29132.25
WiFi Mesh Router 48 29059.94
Kitchen Storage Rack 58 29005.80
Knee Support Brace 63 28785.80
Total unit
Total revenue
Product vs Revenue
60000
50000
40000
30000
20000
10000
0
Values
Total unit
Total revenue
Product_Name
Inserted slicer and connected it to multiple pivot tables using report connections.
Product_Name
Smart LED Light Strip
Soccer Training Con...
Sports Stopwatch B...
Sports Stopwatch Pro
Sports Stopwatch Ti...
Sports Water Hydra...
Sports Wrist Sweat...
Stainless Steel Coo...
Electronics (Category)
Category: North - Electronics
Customer_Dim
Product_Dim
Store_Dim
product vs unit vs revenue
prod category vs unit vs revenu
Ready
Accessibility: Investigate
100%
Search
ENG IN
20:47
10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Calibri 11 Font Alignment General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Add-ins Add-ins
Create a PDF Adobe Acrobat
Elkar Elkar
P13
Product category Sum of units Sum of revenue
Beauty 738 379454.4619
Clothing 844 441461.74
Electronics 1165 592575.61
Home 965 499075.0919
Sports 1241 635440.03
Grand Total 4953 2548106.94
3.Electronics high units sold and high revenue generated compared to clothing
Electronics recorded higher sales due to strong demand for gadgets and frequent product upgrades,
while clothing sales were lower due to varying fashion trends and seasonal purchasing patterns.
Category
Beauty
Clothing
Electronics
Home
Sports
Sum of units Sum of revenue
Product category vs Revenue
25%
15%
17%
19%
24%
Category
Beauty
Clothing
Electronics
Home
Sports
Values
Customer_Dim Product_Dim Store_Dim product vs unit vs revenue prod category vs unit vs revenu
Ready Accessibility: Investigate
100%
Search
ENG IN
20:47
10-10-2026

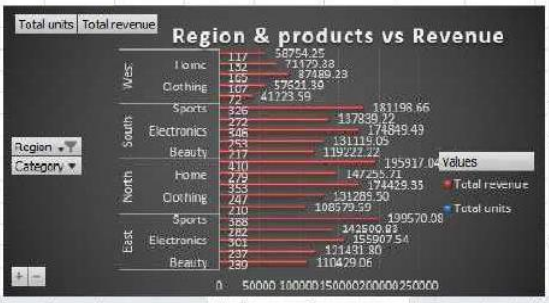

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting Format as Table Cell Styles
Insert Delete Format
Sort & Filter Find & Select
Create a PDF
N3
Store city | Total units | Sum of revenue
Thompsonborough 144 185351.24
Hoodberg 292 147511.15
Bensonhaven 275 143833.83
Williamsfurt 273 125499.18
Sabrinatown 259 138711.99
Port Josephberg 258 134108.91
Port Sharon 255 144009.60
East Davidville 255 129840.12
New Jimmymouth 240 131216.55
East Stevechester 247 122370.55
East Jenniferstad 243 117454.96
Johnsonville 243 136410.49
Vasquezborough 240 109963.71
East Johnstad 236 126549.98
Danielmouth 230 114592.90
Christopherland 224 122885.99
North Nancyview 220 116155.25
Lake Michael 206 109128.64
Washingtonfurt 204 97145.53
New Michaelchester 200 95366.36
1.Bean, Jones and Park store generates the highest overall revenue
2.Thompsonborough performs best in terms of total units sold and revenue contribution?
3.There are no stores have strong sales volume but lower overall revenue efficiency
Total units Sum of revenue
Store vs Revenue
Washingtonfurt
North Nancyview
Danielmouth
Vasquezborough
East Jenniferstad
New Jimmymouth
Port Sharon
Sabrinatown
Bensonhaven
Thompsonborough
0 50000 100000 150000 200000
City
Store_Name
Values
Total units
Sum of revenue
City
East Stevechester
Hoodberg
Johnsonville
Lake Michael
New Jimmymouth
New Michaelchester
North Nancyview
Port Josephberg
Store_Dim
product vs unit vs revenue
prod category vs unit vs revenu
Store city vs unit vs revenue
Regio
Ready
Accessibility: Investigate
100%
Search
ENG IN
20:47 10-10-2026

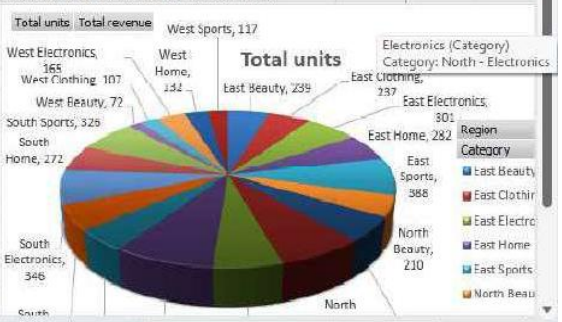
Region & products	Total units	Total revenue
East		
Beauty	239	110429.06
Clothing	237	121431.80
Electronics	301	155907.54
Home	282	142500.83
Sports	388	199570.08
North		
Beauty	210	108579.59
Clothing	247	131289.50
Electronics	353	174429.35
Home	279	147255.71
Sports	410	195917.04
South		
Beauty	217	119222.22
Clothing	253	131119.05
Electronics	346	174849.49
Home	272	137839.22
Sports	326	181198.66
West		
Beauty	72	41223.59

1.The North region achieved the highest revenue due to strong sales volume, followed closely by South and East.
West recorded the lowest revenue, mainly because of its significantly lower unit sales across product categories.

2.The North region contributes the maximum percentage share to total company turnover, driven by its strong sales performance across product categorie

3.Electronics and Sports generated higher revenue due to higher unit sales and the higher value of products sold,
while Beauty recorded lower revenue because fewer units were sold and its overall sales contribution was smaller.





prod category vs unit vs revenu | Store city vs unit vs revenue | Region vs units vs revenue | Cust vs loyalty level

Ready | Accessibility: Investigate

30° | Search | ENG IN | 20:48 10-10-2026

Customer loyalty level	Total revenue
Platinum	
Female	485992.32
Male	283963.25
Bronze	
Female	291791.15
Male	323065.16
Gold	
Female	375937.64
Male	230912.05
Silver	
Female	280967.52
Male	254911.01
Unknown	
Female	5385.06
Male	15181.77
Grand Total	2548106.94

1.Platinum loyalty customers generated the highest total sales.

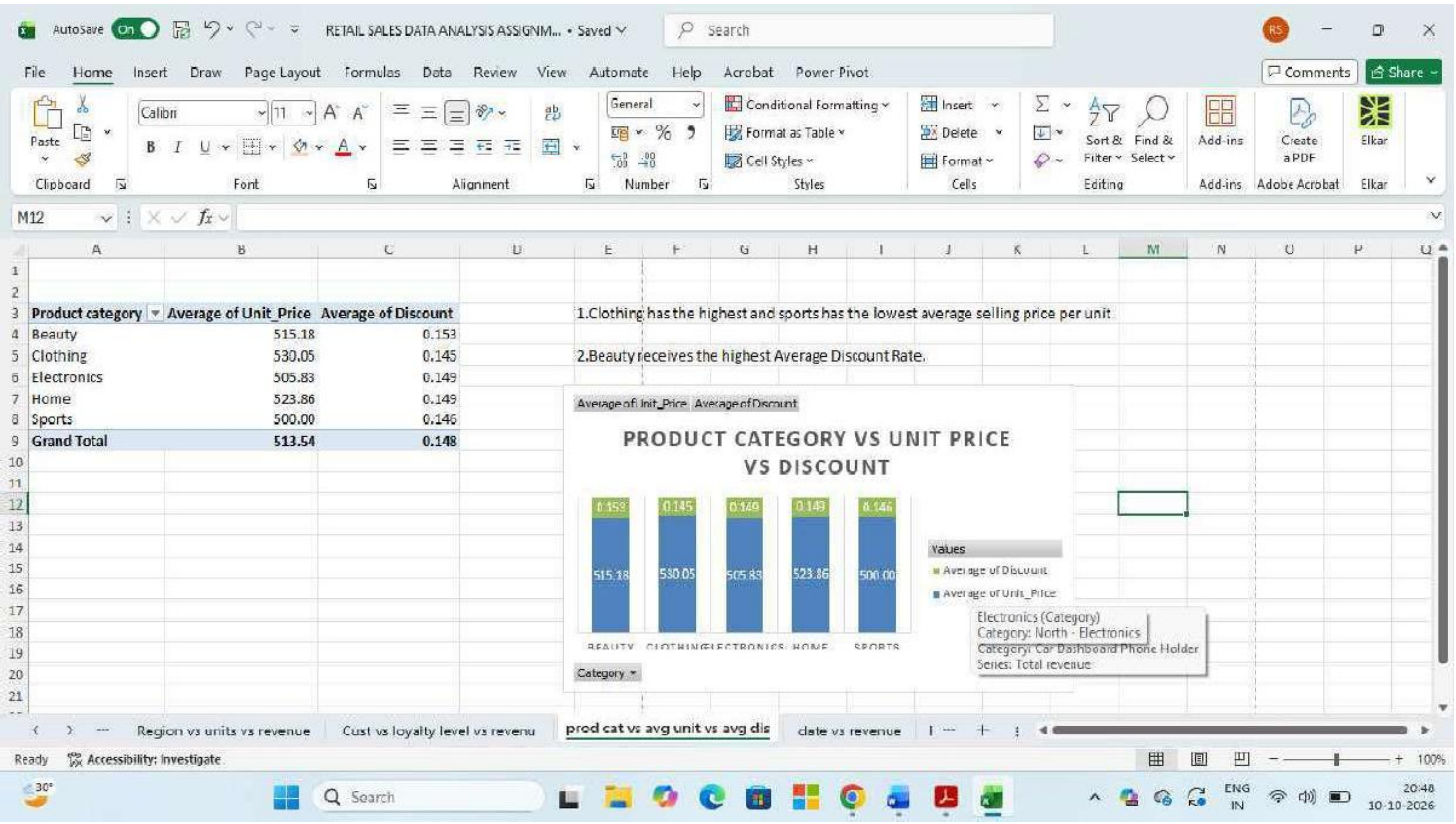


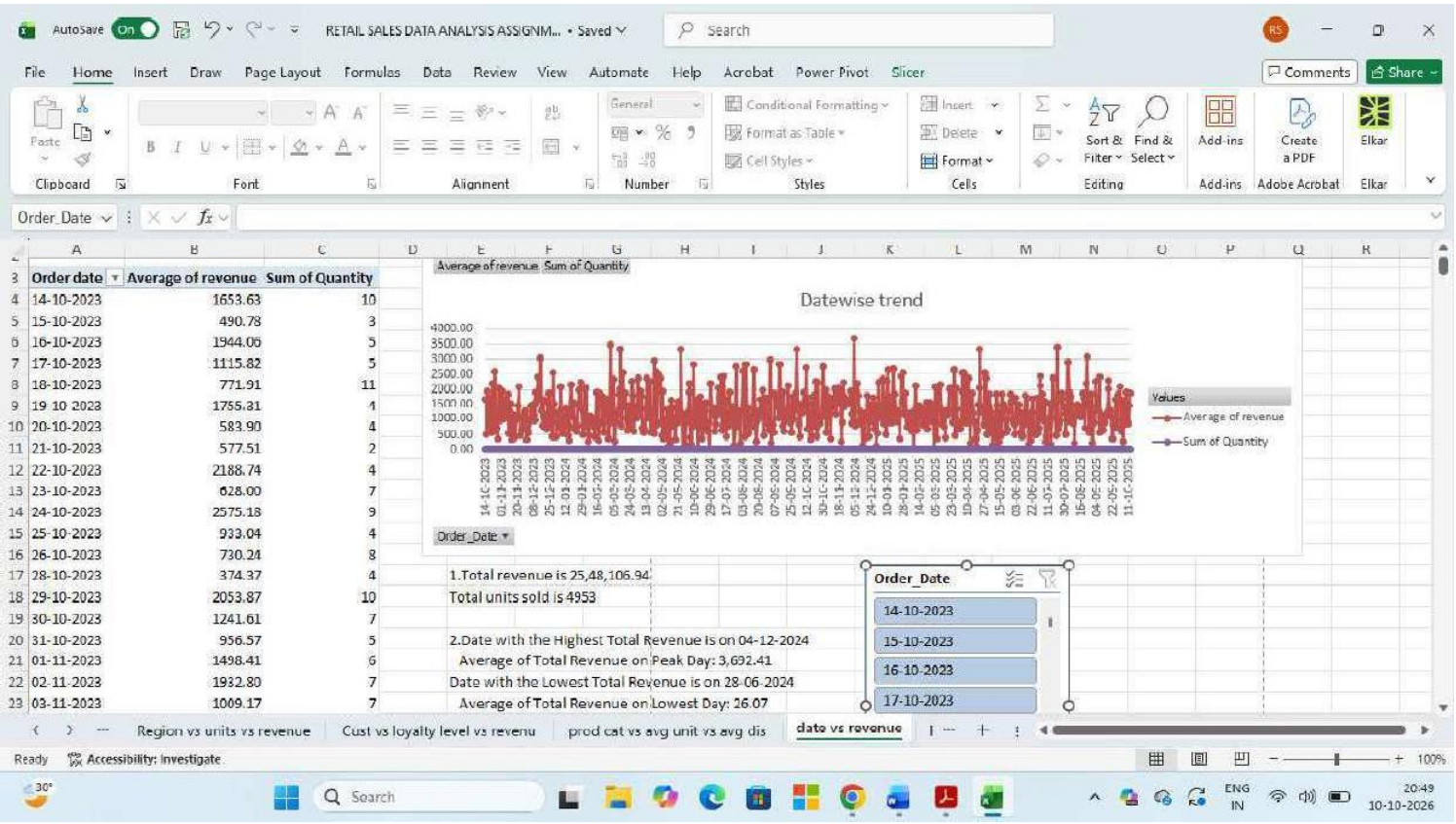


Electronics (Category)
Category: North - Electronics
Category: Car Dashboard Phone Holder
Series: Total revenue

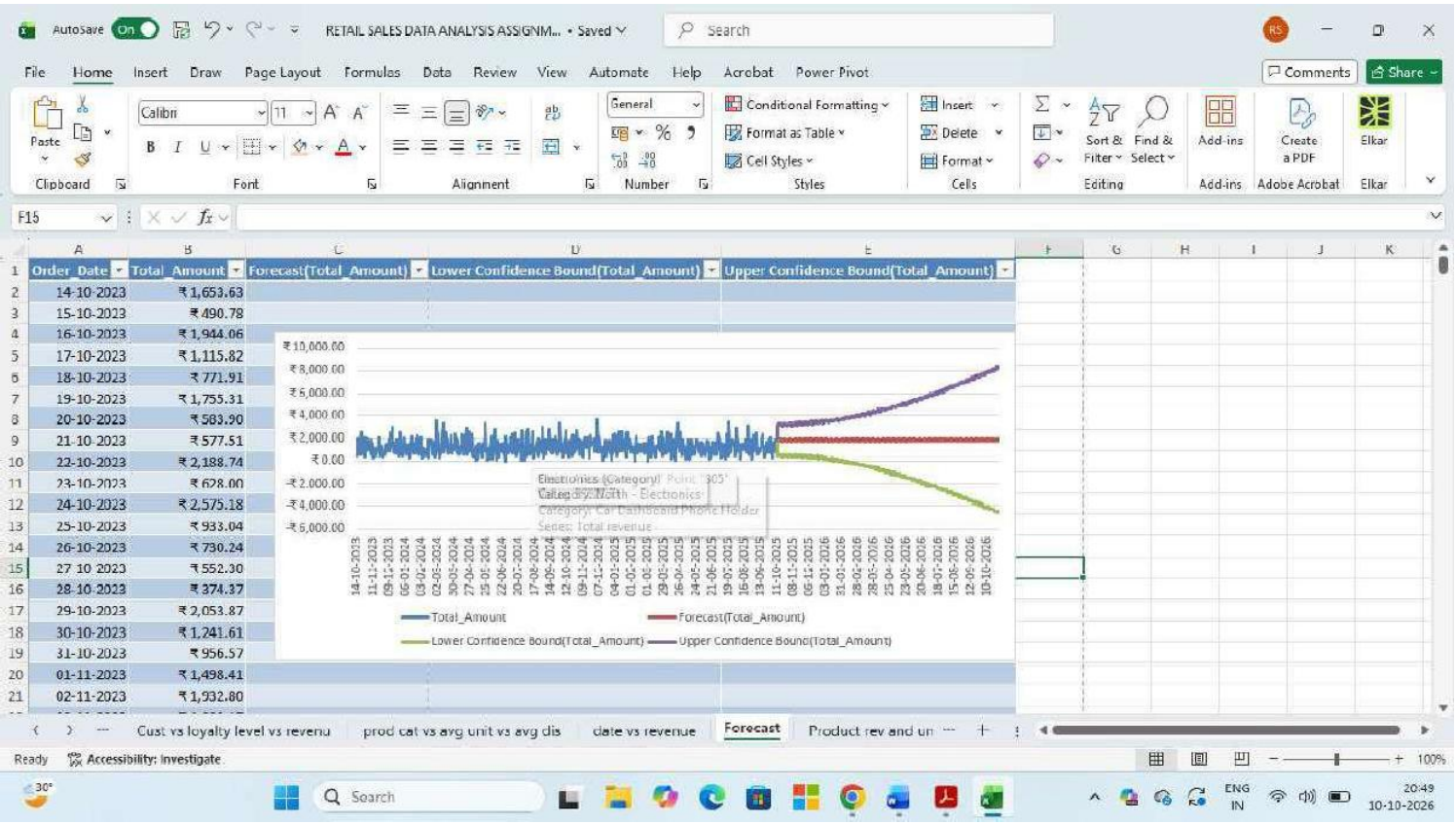
Store city vs unit vs revenue | Region vs units vs revenue | Cust vs loyalty level vs revenu | prod cat vs avg unit vs

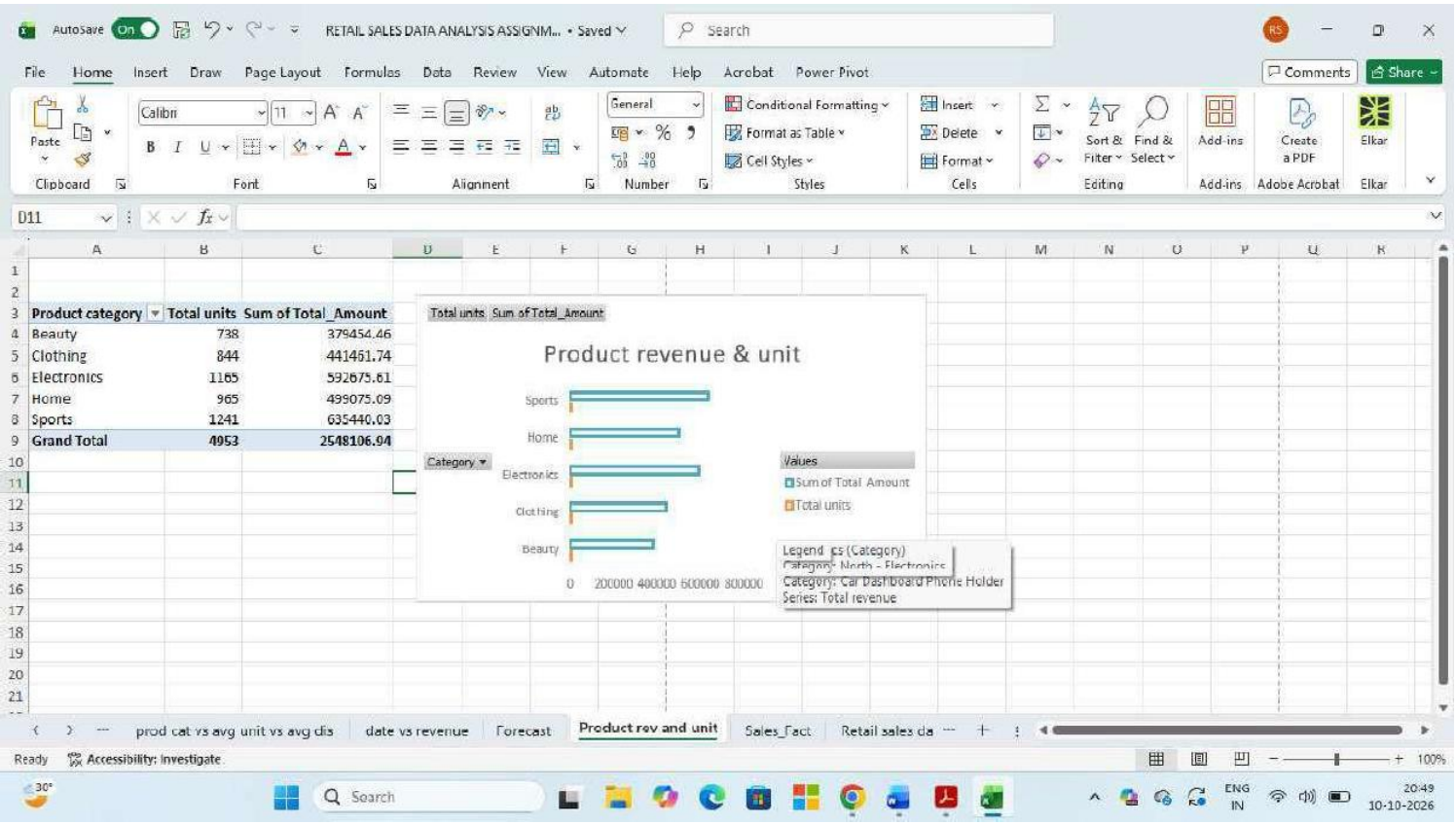
Ready | Accessibility: Investigate | 100%

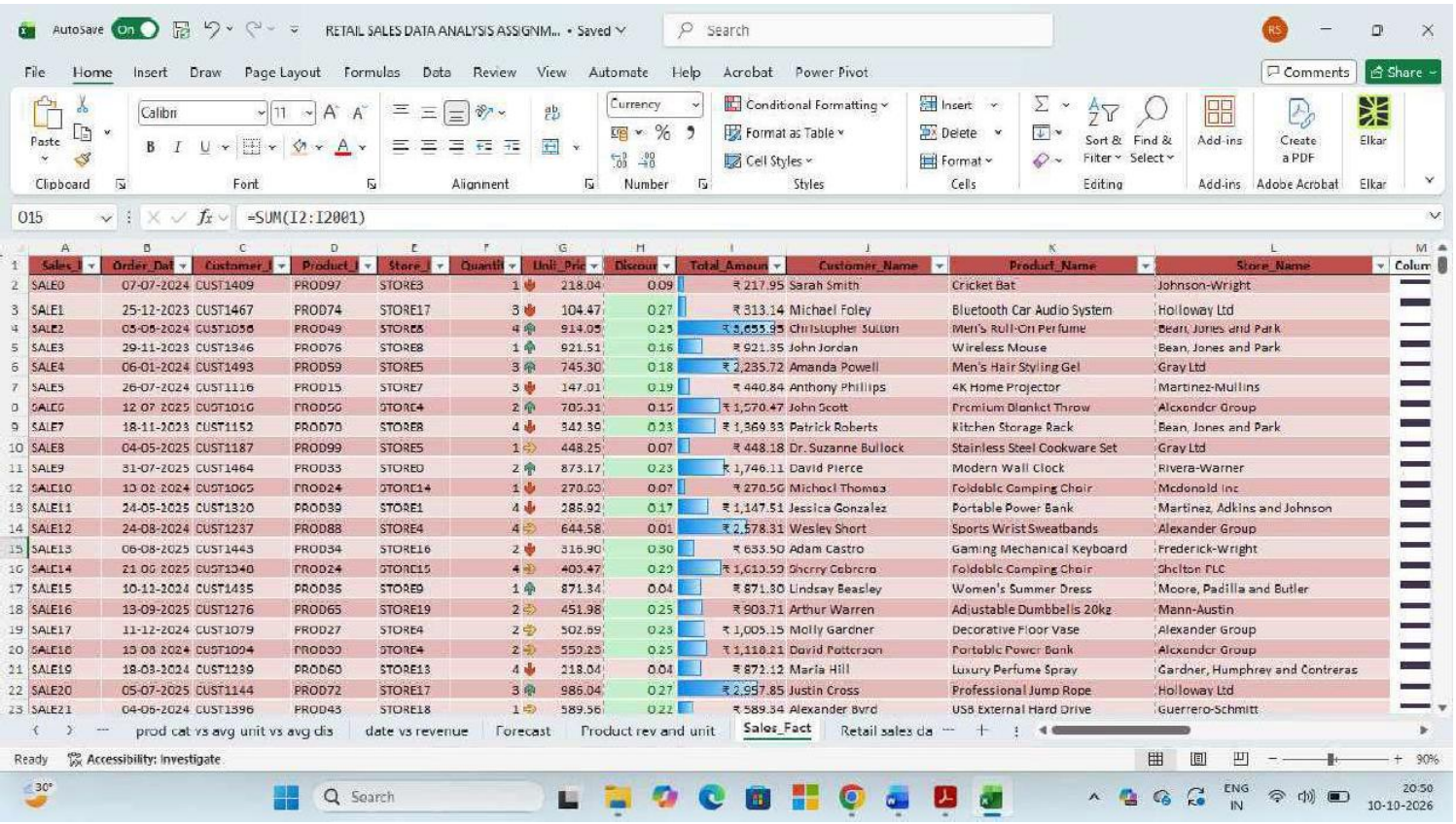

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Calibri 11 General Conditional Formatting Format as Table Cell Styles Insert Delete Format Sort & Filter Find & Select Add-ins Create a PDF Elkar
Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
M12
Product category
Average of Unit_Price
Average of Discount
Beauty 515.18 0.153
Clothing 530.05 0.145
Electronics 505.83 0.149
Home 523.86 0.149
Sports 500.00 0.146
Grand Total 513.54 0.148
1.Clothing has the highest and sports has the lowest average selling price per unit
2.Beauty receives the highest Average Discount Rate.
Average of Unit_Price
Average of Discount
PRODUCT CATEGORY VS UNIT PRICE VS DISCOUNT
0.153
0.145
0.149
0.149
0.146
515.18
530.05
505.83
523.86
500.00
BEAUTY
CLOTHING
ELECTRONICS
HOME
SPORTS
Category
Values
Average of Discount
Average of Unit_Price
Electronics (Category)
Category: North - Electronics
Category: Car Dashboard Phone Holder
Series: Total revenue
Region vs units vs revenue
Cust vs loyalty level vs revenu
prod cat vs avg unit vs avg dis
date vs revenue
Ready
Accessibility: Investigate
100%
Search
ENG IN
20:48
10-10-2026

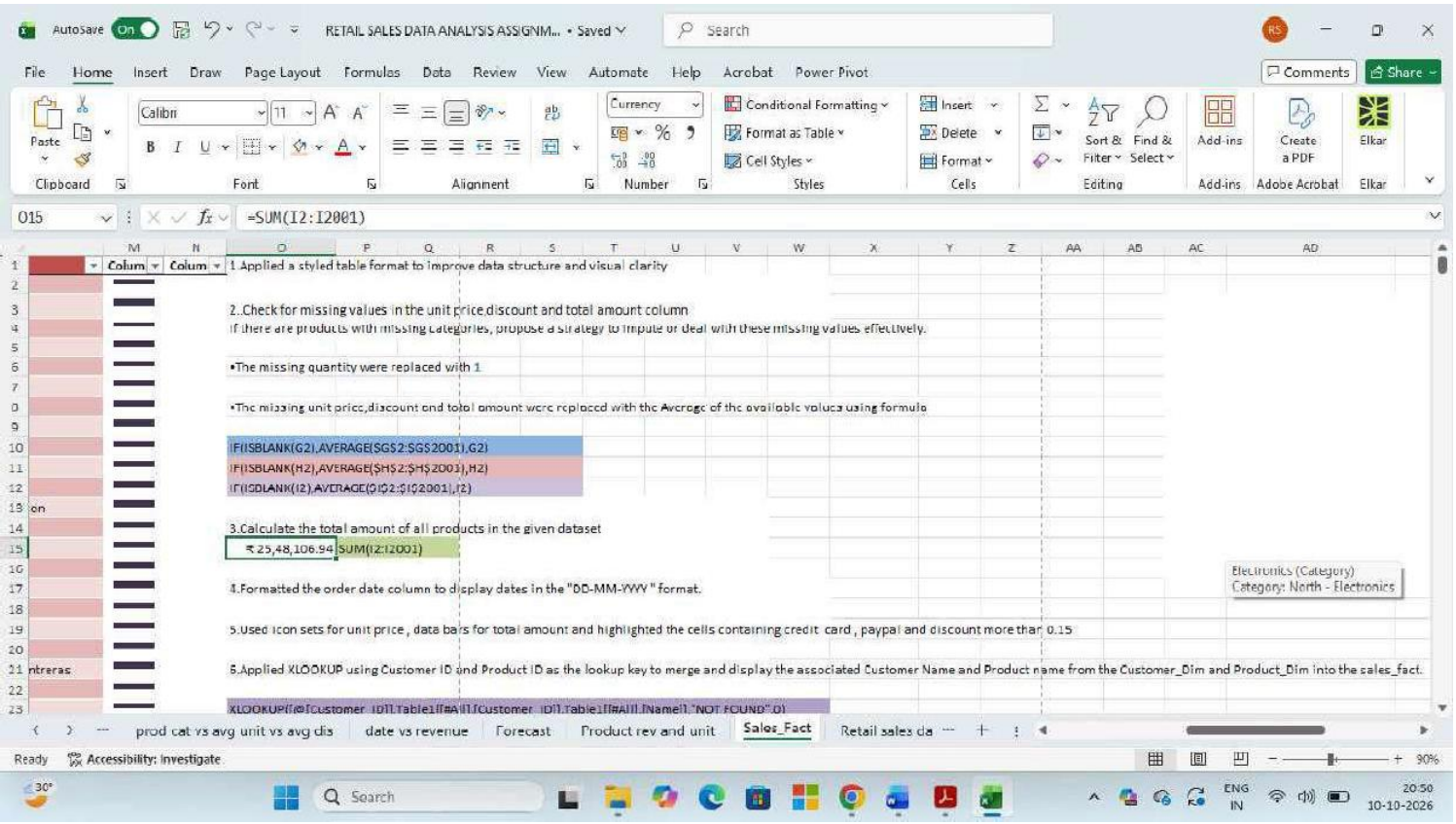


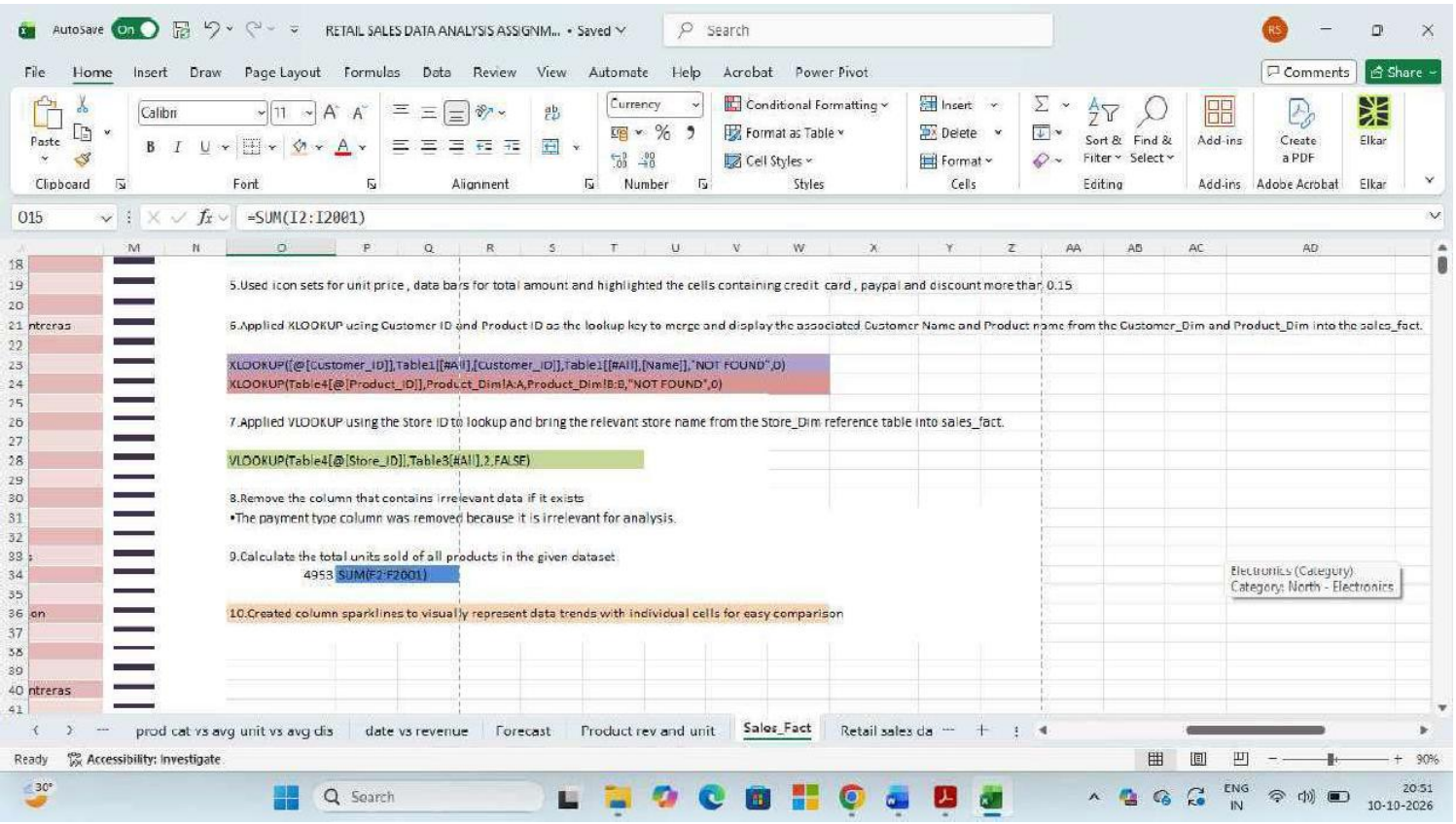
Order date	Average of revenue	Sum of Quantity
14-10-2023	1653.63	10
15-10-2023	490.78	3
16-10-2023	1944.06	5
17-10-2023	1115.82	5
18-10-2023	771.91	11
19-10-2023	1755.81	1
20-10-2023	583.90	4
21-10-2023	577.51	2
22-10-2023	2188.74	4
23-10-2023	628.00	7
24-10-2023	2575.18	9
25-10-2023	933.04	4
26-10-2023	730.24	8
28-10-2023	374.37	4
29-10-2023	2053.87	10
30-10-2023	1241.61	7
31-10-2023	956.57	5
01-11-2023	1498.41	6
02-11-2023	1932.80	7
03-11-2023	1009.17	7

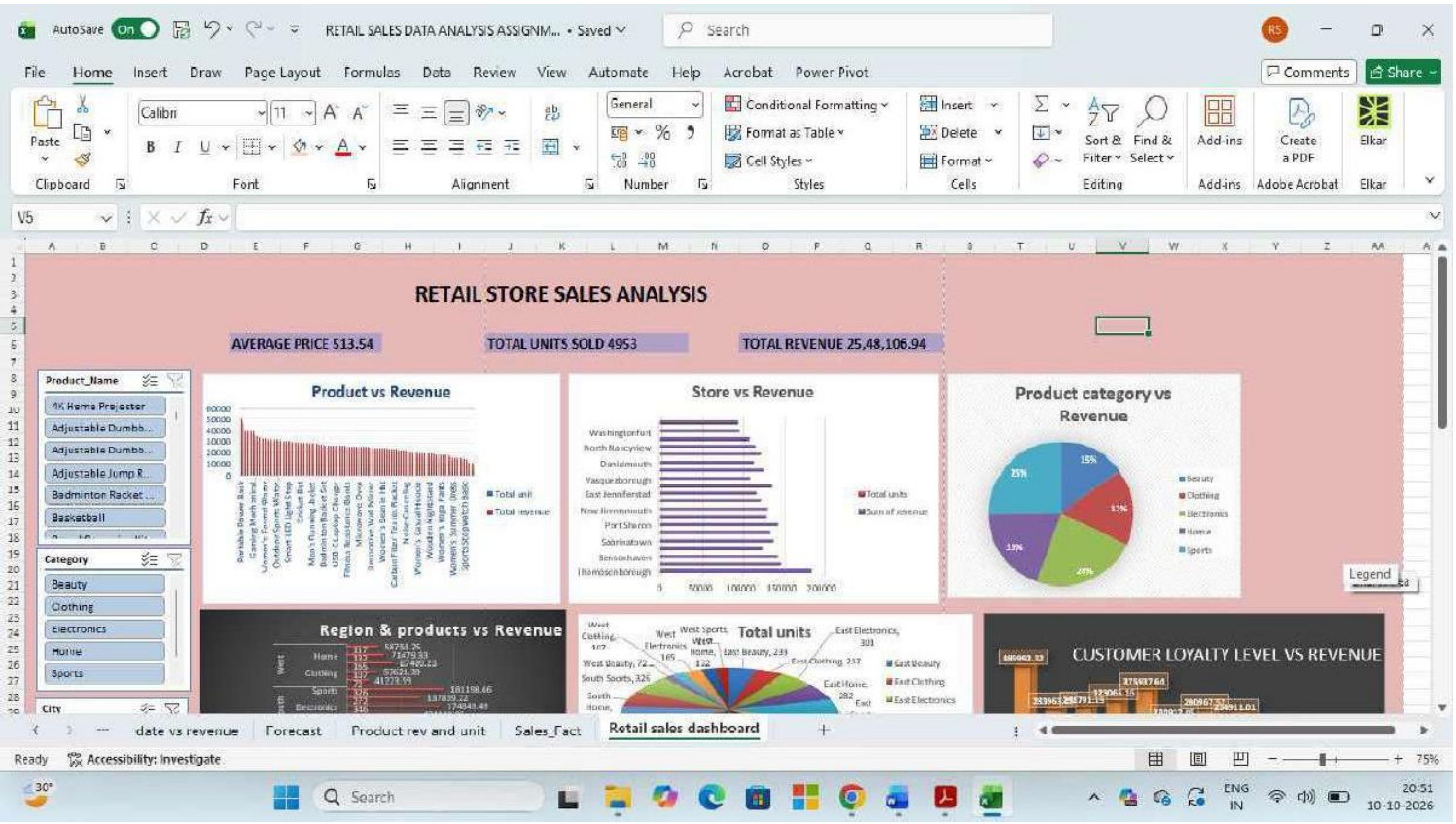

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Sort & Filter
Find & Select
Create a PDF
F15
Order_Date
Total_Amount
Forecast(Total_Amount)
Lower Confidence Bound(Total_Amount)
Upper Confidence Bound(Total_Amount)
14-10-2023 ₹1,653.63
15-10-2023 ₹490.78
16-10-2023 ₹1,944.06
17-10-2023 ₹1,115.82
18-10-2023 ₹771.91
19-10-2023 ₹1,755.31
20-10-2023 ₹583.90
21-10-2023 ₹577.51
22-10-2023 ₹2,188.74
23-10-2023 ₹628.00
24-10-2023 ₹2,575.18
25-10-2023 ₹933.04
26-10-2023 ₹730.24
27-10-2023 ₹552.30
28-10-2023 ₹374.37
29-10-2023 ₹2,053.87
30-10-2023 ₹1,241.61
31-10-2023 ₹956.57
01-11-2023 ₹1,498.41
02-11-2023 ₹1,932.80
₹10,000.00
₹8,000.00
₹6,000.00
₹4,000.00
₹2,000.00
₹0.00
-₹2,000.00
-₹4,000.00
-₹6,000.00
Total_Amount
Forecast(Total_Amount)
Lower Confidence Bound(Total_Amount)
Upper Confidence Bound(Total_Amount)
Cust vs loyalty level vs revenu
prod cat vs avg unit vs avg dis
date vs revenue
Forecast
Product rev and un
Ready
Accessibility: Investigate
100%
Search
ENG IN
20:49
10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Calibri 11 Font Alignment General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Add-ins Add-ins
Create a PDF Adobe Acrobat
Elkar Elkar
D11
Product category Total units Sum of Total_Amount
Beauty 738 379454.46
Clothing 844 441451.74
Electronics 1165 592675.61
Home 965 499075.09
Sports 1241 635440.03
Grand Total 4953 2548106.94
Total units Sum of Total_Amount
Product revenue & unit
Sports
Home
Electronics
Clothing
Beauty
Category
0 200000 400000 600000 800000
Values
Sum of Total_Amount
Total units
Legend (Series) (Category)
Category: North - Electronics
Category: Car Dashboard Phone Holder
Series: Total revenue
prod cat vs avg unit vs avg dis
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales da
Ready
Accessibility: Investigate
100%
Search
ENG IN
20:49 10-10-2026

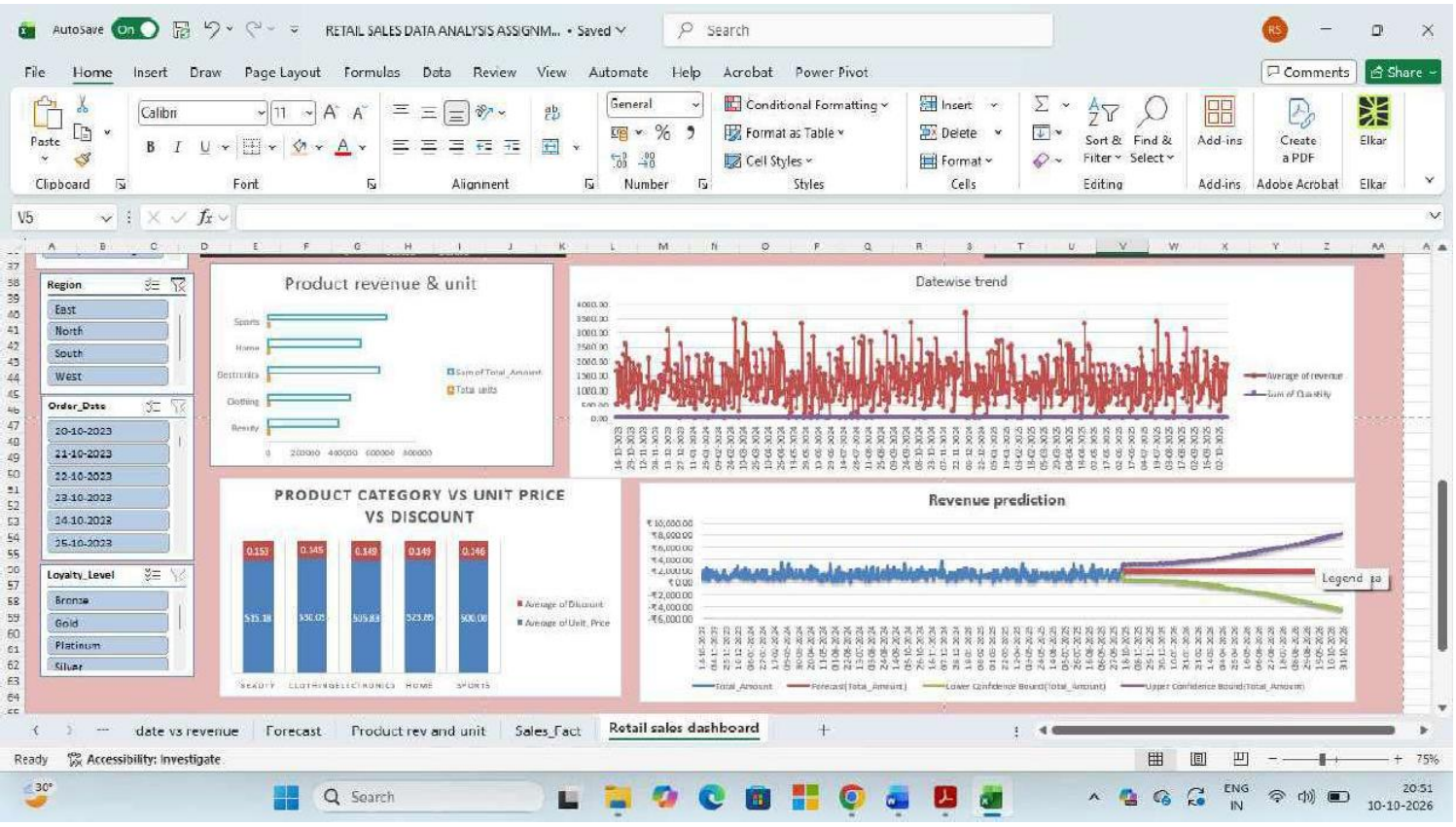

RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Sort & Filter
Find & Select
Create a PDF
O15
=SUM(I2:I2001)
Sales_I Order_Dat Customer_I Product_I Store_I Quantit Unit_Pric Discoun Total_Amoun Customer_Name Product_Name Store_Name Colum
SALE0 07-07-2024 CUST1409 PROD97 STORE3 1 218.04 0.09 ₹ 217.95 Sarah Smith Cricket Bat Johnson-Wright
SALE1 25-12-2023 CUST1467 PROD74 STORE17 3 104.47 0.27 ₹ 313.14 Michael Foley Bluetooth Car Audio System Holloway Ltd
SALE2 05-05-2024 CUST1056 PROD49 STORE6 4 914.05 0.25 ₹ 3,655.95 Christopher Sutton Men's Roll-On Perfume Bean, Jones and Park
SALE3 29-11-2023 CUST1346 PROD76 STORE8 1 921.51 0.16 ₹ 921.35 John Jordan Wireless Mouse Bean, Jones and Park
SALE4 06-01-2024 CUST1493 PROD59 STORE5 3 745.30 0.18 ₹ 2,235.72 Amanda Powell Men's Hair Styling Gel Gray Ltd
SALE5 26-07-2024 CUST1116 PROD15 STORE7 3 147.01 0.19 ₹ 440.84 Anthony Phillips 4K Home Projector Martinez-Mullins
SALE6 12-07-2025 CUST1016 PROD56 STORE4 2 785.31 0.15 ₹ 1,570.47 John Scott Premium Blanket Throw Alexander Group
SALE7 18-11-2023 CUST1152 PROD70 STORE8 4 342.30 0.23 ₹ 1,369.33 Patrick Roberts Kitchen Storage Rack Bean, Jones and Park
SALE8 04-05-2025 CUST1187 PROD99 STORE5 1 448.25 0.07 ₹ 448.18 Dr. Suzanne Bullock Stainless Steel Cookware Set Gray Ltd
SALE9 31-07-2025 CUST1464 PROD33 STORE0 2 873.17 0.23 ₹ 1,746.11 David Pierce Modern Wall Clock Rivera-Warner
SALE10 13-02-2024 CUST1065 PROD24 STORE14 1 270.60 0.07 ₹ 270.56 Michael Thomas Foldable Camping Chair Mcdonald Inc
SALE11 24-05-2025 CUST1320 PROD39 STORE1 4 285.92 0.17 ₹ 1,147.51 Jessica Gonzalez Portable Power Bank Martinez, Adkins and Johnson
SALE12 24-08-2024 CUST1237 PROD88 STORE4 4 644.58 0.01 ₹ 2,578.31 Wesley Short Sports Wrist Sweatbands Alexander Group
SALE13 06-08-2025 CUST1443 PROD34 STORE16 2 316.90 0.30 ₹ 633.50 Adam Castro Gaming Mechanical Keyboard Frederick-Wright
SALE14 21-06-2025 CUST1040 PROD24 STORE15 4 403.47 0.25 ₹ 1,610.59 Sherry Cabrera Foldable Camping Chair Shelton PLC
SALE15 10-12-2024 CUST1435 PROD35 STORE9 1 871.34 0.04 ₹ 871.30 Lindsay Beasley Women's Summer Dress Moore, Padilla and Butler
SALE16 13-09-2025 CUST1276 PROD65 STORE19 2 451.98 0.25 ₹ 903.71 Arthur Warren Adjustable Dumbbells 20kg Mann-Austin
SALE17 11-12-2024 CUST1079 PROD27 STORE4 2 502.69 0.23 ₹ 1,005.15 Molly Gardner Decorative Floor Vase Alexander Group
SALE18 13-08-2024 CUST1004 PROD30 STORE4 2 559.25 0.25 ₹ 1,118.21 David Patterson Portable Power Bank Alexander Group
SALE19 18-03-2024 CUST1239 PROD60 STORE13 4 218.04 0.04 ₹ 871.12 Maria Hill Luxury Perfume Spray Gardner, Humphrey and Contreras
SALE20 05-07-2025 CUST1144 PROD72 STORE17 3 986.04 0.27 ₹ 2,957.85 Justin Cross Professional Jump Rope Holloway Ltd
SALE21 04-06-2024 CUST1396 PROD43 STORE18 1 589.56 0.22 ₹ 589.34 Alexander Byrd USB External Hard Drive Guerrero-Schmitt
prod cat vs avg unit vs avg dis
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales da
Ready
Accessibility: Investigate
90%
Search
ENG IN
20:50 10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Calibri 11 Currency Conditional Formatting Format as Table Cell Styles Insert Delete Format Sort & Filter Find & Select Add-ins Create a PDF Elkar
Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
O15 =SUM(I2:I2001)
Colum Colum
1 Applied a styled table format to improve data structure and visual clarity
2..Check for missing values in the unit price,discount and total amount column
If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.
•The missing quantity were replaced with 1.
•The missing unit price,discount and total amount were replaced with the Average of the available values using formula
IF(ISBLANK(G2),AVERAGE($G$2:$G$2001),G2)
IF(ISBLANK(H2),AVERAGE($H$2:$H$2001),H2)
IF(ISBLANK(I2),AVERAGE($I$2:$I$2001),I2)
3.Calculate the total amount of all products in the given dataset
₹ 25,48,106.94 SUM(I2:I2001)
4.Formatted the order date column to display dates in the "DD-MM-YYYY " format.
Electronics (Category)
Category: North - Electronics
5.Used icon sets for unit price , data bars for total amount and highlighted the cells containing credit card , paypal and discount more than 0.15
6.Applied XLOOKUP using Customer ID and Product ID as the lookup key to merge and display the associated Customer Name and Product name from the Customer_Dim and Product_Dim into the sales_fact.
XLOOKUP(I2,Customer_ID[1],Table1[#All],Customer_ID[1],Table1[#All],Name[],"NOT FOUND",0)
prod cat vs avg unit vs avg dis
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales da
Ready
Accessibility: Investigate
90%
Search
ENG IN
20:50 10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Calibri 11 Currency Conditional Formatting Format as Table Cell Styles Insert Delete Format Sort & Filter Find & Select Add-ins Create a PDF Elkar
Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
O15 =SUM(I2:I2001)
5.Used icon sets for unit price , data bars for total amount and highlighted the cells containing credit card , paypal and discount more than 0.15
6.Applied XLOOKUP using Customer ID and Product ID as the lookup key to merge and display the associated Customer Name and Product name from the Customer_Dim and Product_Dim into the sales_fact.
XLOOKUP([@[Customer_ID]],Table1[[#All],[Customer_ID]],Table1[[#All],[Name]],"NOT FOUND",0)
XLOOKUP(Table4[@[Product_ID]],Product_Dim!A:A,Product_Dim!B:B,"NOT FOUND",0)
7.Applied VLOOKUP using the Store ID to lookup and bring the relevant store name from the Store_Dim reference table into sales_fact.
VLOOKUP(Table4[@[Store_ID]],Table3[#All],2,FALSE)
8.Remove the column that contains irrelevant data if it exists
•The payment type column was removed because it is irrelevant for analysis.
9.Calculate the total units sold of all products in the given dataset
4953 SUM(F2:F2001)
Electronics (Category)
Category: North - Electronics
10.Created column sparklines to visually represent data trends with individual cells for easy comparison
prod cat vs avg unit vs avg dis
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales da
Ready
Accessibility: Investigate
90%
Search
ENG IN
20:51 10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments Share
Paste Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat Elkar
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Sort & Filter
Find & Select
Create a PDF
V5
RETAIL STORE SALES ANALYSIS
AVERAGE PRICE $13.54
TOTAL UNITS SOLD 4953
TOTAL REVENUE 25,48,106.94
Product_Name
4K Home Projector
Adjustable Dumbb...
Adjustable Dumbb...
Adjustable Jump R...
Badminton Racket ...
Basketball
Category
Beauty
Clothing
Electronics
Home
Sports
City
Product vs Revenue
Total unit
Total revenue
Store vs Revenue
Total units
Sum of revenue
Product category vs Revenue
Beauty
Clothing
Electronics
Home
Sports
Legend
Region & products vs Revenue
Total units
East Beauty
East Clothing
East Electronics
CUSTOMER LOYALTY LEVEL VS REVENUE
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales dashboard
Ready
Accessibility: Investigate
75%
Search
ENG IN
20:51
10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments
Share
Clipboard
Font
Alignment
Number
Styles
Cells
Editing
Add-ins
Adobe Acrobat
Elkar
Region & products vs Revenue
Total units
CUSTOMER LOYALTY LEVEL VS REVENUE
Product revenue & unit
Datewise trend
Clothing
Electronics
Home
Sports
City
North Nancyview
Port Josephberg
Port Sharon
Sabrinatown
Thompsonborough
Region
East
North
South
West
Order Date
20-10-2023
21-10-2023
Total revenue
Total units
Sum of Total_Amount
Total units
Average of revenue
Sum of Quantity
Chart Area
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales dashboard
Ready
Accessibility: Investigate
75%
Search
ENG
IN
20:51
10-10-2026


RETAIL SALES DATA ANALYSIS ASSIGNM... • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot
Comments
Share
Region
East
North
South
West
Order_Date
20-10-2023
21-10-2023
22-10-2023
23-10-2023
24-10-2023
25-10-2023
Loyalty_Level
Bronze
Gold
Platinum
Silver
Product revenue & unit
Sports
Home
Electronics
Clothing
Beauty
Sum of Total_Amount
Total Units
Datewise trend
Average of revenue
Sum of Quantity
PRODUCT CATEGORY VS UNIT PRICE VS DISCOUNT
0.153
0.145
0.149
0.149
0.146
Average of Discount
Average of Unit_Price
BEAUTY
CLOTHING
ELECTRONICS
HOME
SPORTS
Revenue prediction
Legend
Total_Amount
Forecast(Total_Amount)
Lower Confidence Bound(Total_Amount)
Upper Confidence Bound(Total_Amount)
date vs revenue
Forecast
Product rev and unit
Sales_Fact
Retail sales dashboard
Ready
Accessibility: Investigate
75%
Search
ENG
IN
20:51
10-10-2026

EXCEL PROJECT: E-COMMERCE SALES PERFORMANCE & PIVOT TABLE ANALYSIS

Problem Statement

A retail enterprise operates across multiple regional branches (East, North, South, and West), offering merchandise across diverse categories such as Electronics, Clothing, Home, and Sports. Management requires an analysis of sales transaction data from late 2023 through late 2025 to evaluate revenue momentum, geographic market share, and customer buying behavior.

The raw data is organized across relational tables (Sales_Fact, Customer_Dim, Product_Dim, and Store_Dim.

To support executive decision-making, you are required to analyze the transactional data using Pivot Tables, Pivot Charts, Data Modeling (Lookups), and an Interactive Dashboard to transform raw records into actionable insights.

🎯 Objectives of the Excel Analysis

The goal of this project is to:

- **Summarize large datasets using Pivot Tables.**
- **Visualize trends and category distributions using Pivot Charts.**
- **Build relational lookups and calculations using Excel formulas (XLOOKUP / VLOOKUP).**
- **Create a dynamic and interactive dashboard with linked Slicers and Timelines.**
- **Present business findings and strategic recommendations for management review.**

📌 Business Questions to Be Answered

1. Overall Sales Performance

- **What is the total revenue and total units sold?**
- **Which date recorded the highest and lowest total revenue?**

2. Product & Category Performance Analysis

- **Which Product contributes the highest overall revenue and units sold?**
- **Which products drive high sales volume (Quantity) but yield lower total revenue?**
- **Compare performance between electronics and clothing.**

3. Store Performance

- **Which store generates the highest overall revenue?**
- **Which City performs best in terms of total units sold and revenue contribution?**
- **Are there stores with strong sales volume but lower overall revenue efficiency?**

4. Regional Analysis

- **How does sales revenue compare across the four geographic Regions (East, North, South, and West)?**
- **Which region contributes the maximum percentage share to total company turnover?**
- **How do product category preferences (Category) vary across each regional territory?**

5. Customer Performance Analysis

- **Which customer Loyalty Level (Bronze, Silver, Gold, Platinum) generates the highest total sales?**
- **Who spends more in total revenue: Male or Female customers?**

6. Pricing & Revenue Efficiency

Pricing & Revenue Efficiency

- **Which product category has the highest and lowest average selling price per unit (Unit_Price)?**
- **Which category receives the highest Average Discount Rate?**

RETAIL SALES DATA ANALYSIS

Four Types of Business Analytics

1. Descriptive Analytics – What Happened?

- **Total Revenue: ₹2,548,106.94**
- **Total Units Sold: 4,953 units**
- **Top Product by Revenue: Portable Power Bank – ₹50,615.34**
- **Top Product by Quantity: Men's Casual Sneakers – 87 units**
- **Top Category: Sports – ₹635,440.03**
- **Top Region: North – ₹757,471.19**
- **Top Store: Bean, Jones and Park – ₹185,351.24**
- **Top Loyalty Tier: Platinum – ₹769,955.57**

Analysis: Sports generated the highest revenue. North was the best-performing region, and Platinum customers contributed the most revenue.

2. Diagnostic Analytics – Why Did It Happen?

- **Sports sold more units than Electronics.**
- **West recorded fewer transactions than North.**
- **Platinum customers had the highest number of transactions.**

Analysis: Differences in the number of units sold and transactions contributed to the differences in revenue. More information is needed to identify the exact reasons.

3. Predictive Analytics – What Could Happen Next?

- **The Forecast sheet estimates future sales using past data.**
- **The forecast results are uncertain and may not be accurate.**
- **Sports products and Portable Power Banks can be monitored for future demand.**

Analysis: Past sales can help estimate future demand, but actual sales may change.

4. Prescriptive Analytics – What Should Be Done?

- **Maintain sufficient stock of popular products.**
- **Investigate the reasons for lower sales in the West region.**
- **Provide suitable offers to retain Platinum customers.**
- **Check product costs before deciding which category is most profitable.**
- **Verify the discount calculation before reporting discount percentages.**

Analysis: Management should improve stock planning, investigate weak sales areas, and use accurate data to make business decisions.

Conclusion

Sports was the highest-revenue category, North was the leading region, and Platinum customers generated the highest revenue. The business should focus on popular products, improve sales in the West region, and use reliable forecasts for future planning.